Naïve Bayes

From Evidence to Intelligent Predictions

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How can a machine update its belief using evidence?

An interactive journey from probability to classification

A Doctor's Puzzle

A patient has a fever.

Does that mean the patient has COVID-19?

What we already know

- COVID-19 is rare
- Fever is common

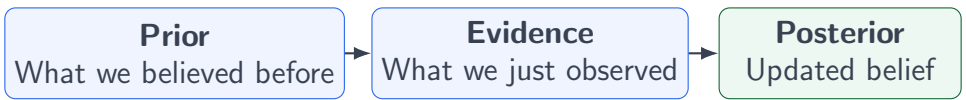
New evidence

Fever observed!

Challenge

Should our belief about COVID-19 increase?

How Should Our Belief Change?

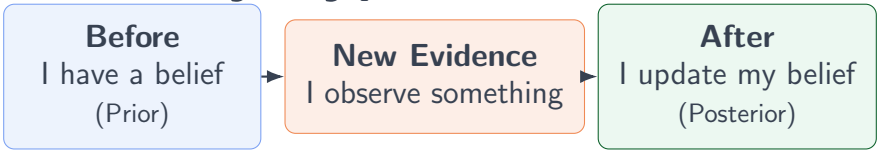


Bayesian learning = updating belief using evidence

New Evidence Can Change What We Believe

Suppose we are trying to answer:

Is my hypothesis true?



The Central Idea

What I believed before + What I observe now → What I should believe next

Bayes' Theorem

$$P(H|E) = \frac{P(E|H)\,P(H)}{P(E)}$$

Posterior	**Likelihood**	**Prior**	**Evidence**
What we want to know	Evidence if hypothesis is true	What we knew before	Normalization

The big idea

$$\textbf{Posterior} = \textbf{Prior} \times \textbf{Evidence}$$

From Bayes to Machine Learning

Instead of asking:

"Is hypothesis H true?"

We ask:

"Which class is most probable?"

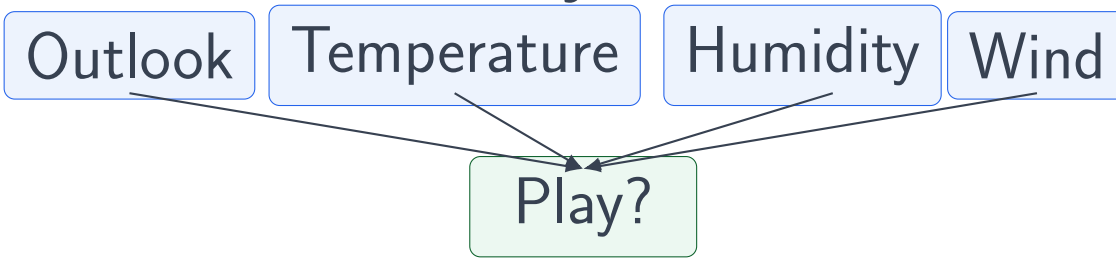
Class A **Class B**

Choose the class with larger posterior probability

The Problem: Too Many Features!

Suppose we want to classify:

Will we Play Tennis?



Challenge

How do we estimate probabilities for every possible combination?

The Naïve Shortcut

What if we assume...

Once we know the class,

the features act independently?

Know the class $\Rightarrow$ treat features separately

This is why it is called *Naïve*

The assumption is often unrealistic — but surprisingly useful.

Breaking a Big Probability into Small Ones

Instead of estimating:

$$P(X_1, X_2, X_3, X_4 \mid C)$$

Naïve Bayes assumes:

$$P(X_1, X_2, X_3, X_4 \mid C) \approx P(X_1|C)P(X_2|C)P(X_3|C)P(X_4|C)$$

Why is this powerful?

One difficult probability becomes several easy probabilities.

Naïve Bayes in One Equation

$$P(C|X) \propto P(C) \prod_i P(x_i|C)$$

Prior	$\times$	**Evidence from each feature**
$P(C)$	$\times$	$\prod_i P(x_i\|C)$

Prediction Rule

Calculate a score for every class.
Choose the largest one.

Naïve Bayes Classifier Example: Training Dataset

Class

- play golf = **yes**
- play golf = **no**

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Test Instance

Outlook	Temp	Humidity	Windy	Play
Rainy	Cool	High	True	?

Prediction Game

A new day arrives:

Outlook	Temperature	Humidity	Wind
Rainy	Cool	High	True

Play Tennis?

YES or NO

Don't guess randomly!

Which class receives more probabilistic support?

Step 1: What Did We Know Before Seeing Today?

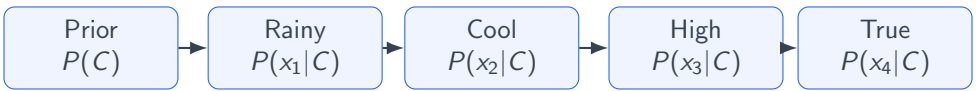
Suppose the training dataset contains:

Class	Count	Prior
Play = Yes	9	9/14
Play = No	5	5/14

Before seeing today's weather, YES is slightly more likely.

Step 2: Let the Evidence Speak

For each possible class, calculate:



Multiply the evidence!

Naïve Bayes Classifier Example: Likelihood Tables

Frequency Table

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3

Likelihood Table

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Temp.	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Windy	False	6/9	2/5
	True	3/9	3/5

Naïve Bayes Classifier Example: Classification

Outlook	Temp	Humidity	Windy	Play
Rainy	Cool	High	True	?

$$P(Yes \mid X) = P(Rainy \mid Yes) \times P(Cool \mid Yes) \times P(High \mid Yes) \times P(True \mid Yes) \times P(Yes) \;/\; P(x)$$

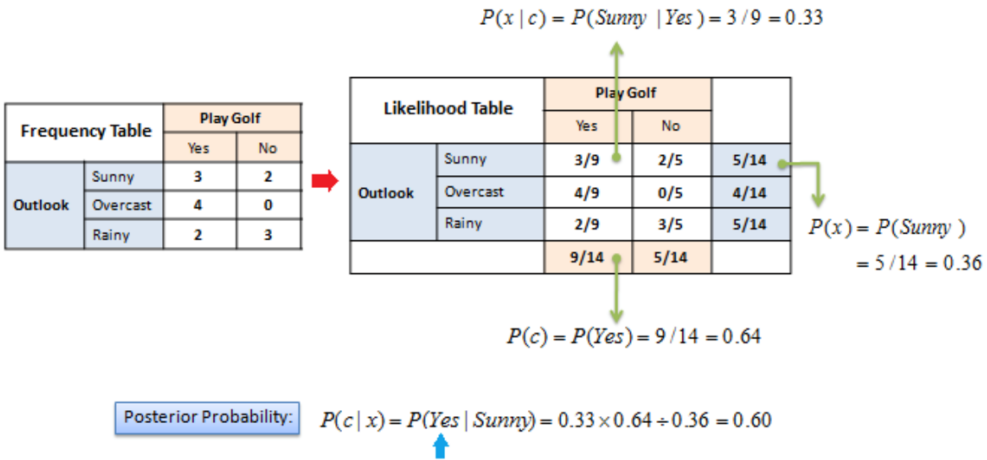
$$P(Yes \mid X) = \underbrace{2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14}_{P(x)} = 0.00529 \rightarrow 0.2 = \frac{0.00529}{0.02057 + 0.00529}$$

$$P(No \mid X) = P(Rainy \mid No) \times P(Cool \mid No) \times P(High \mid No) \times P(True \mid No) \times P(No) \;/\; P(x)$$

$$P(No \mid X) = \underbrace{3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14}_{P(x)} = 0.02057 \rightarrow 0.8 = \frac{0.02057}{0.02057 + 0.00529}$$

$P(x) = P(x \mid yes) * P(yes) + P(x \mid no) * P(no)$

Naïve Bayes Classifier Example: $P(\textbf{Yes} \mid \textbf{Sunny})$



Naïve Bayes Classifier Example: $P(\text{No} \mid \text{Sunny})$

$$P(x \mid c) = P(Sunny \mid No) = 2/5 = 0.4$$

Frequency Table		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3

		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
		9	5	14

$$P(x) = P(Sunny) = 5/14 = 0.36$$

$$P(c) = P(No) = 5/14 = 0.36$$

Posterior Probability: $P(c \mid x) = P(No \mid Sunny) = 0.40 \times 0.36 \div 0.36 = 0.40$

The Zero Probability Trap

What happens if...

$$P(X_1|C) = 0$$

Then:

$$P(C|X) \propto P(C) \times \boxed{0} \times P(X_2|C) \times P(X_3|C)$$

$$= 0$$

One unseen feature can destroy the entire prediction!

A Small Problem That Can Break Naïve Bayes

Suppose the model has seen 1000 training examples.

Income	Number of Examples
Low	0
Medium	990
High	10

What happens if a new person has:

$$\boxed{\text{Income} = \text{Low}}$$

The Problem: Zero Probability

The model estimates probability using frequency:

$$P(\text{Income} = \text{Low}) = \frac{0}{1000} = \boxed{0}$$

But Naïve Bayes multiplies probabilities:

$$P(C|X) \propto P(C) \times P(x_1|C) \times \boxed{0} \times P(x_3|C)$$

Everything becomes **0**!

But Does "Never Seen" Mean "Impossible"?

No!

The training data may simply be incomplete.

Never observed in training data $\neq$ Impossible in real life

Key Idea

We need to give unseen events a small probability instead of zero.

The Solution: Add a Small Safety Net

Laplace Smoothing

Instead of using the original count:

$$\frac{\text{count}}{N}$$

We add 1 to every possible value:

$$\frac{\text{count} + 1}{N + k}$$

Why add k to the denominator?

Let's Apply Laplace Smoothing

Original counts:

Income	Count
Low	0
Medium	990
High	10

Add 1 to every category:

Income	New Count
Low	1
Medium	991
High	11

Notice

The total also changes from:

$$1000 \rightarrow 1003$$

Now No Event Has Zero Probability

$$N = 1000, \qquad k = 3$$

$$N + k = 1003$$

Income	Probability after smoothing
Low	$\frac{1}{1003}$
Medium	$\frac{991}{1003}$
High	$\frac{11}{1003}$

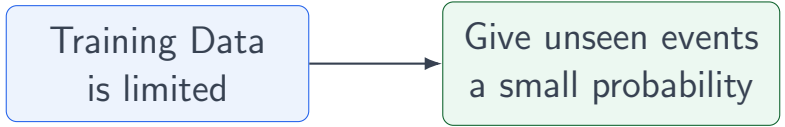
Zero is gone!

The One Idea to Remember

Never Seen

does not mean

Impossible



Can Naïve Bayes Handle Different Types of Data?

A feature can be represented in different ways.

Categorical
Sunny, Rainy, Cloudy

Continuous
23.5, 48.2, 72.1

The Question

How should Naïve Bayes calculate probability in each case?

Categorical Features: Start by Counting

Suppose we want to calculate:

$$P(\text{Sunny} \mid \text{Play} = \text{Yes})$$

Among Play = Yes	**Count**
Sunny	3
Other Outlook values	6
Total	9

$$P(\text{Sunny} \mid \text{Yes}) = \frac{3}{9} = 0.33$$

But What About Continuous Values?

For categorical data, we can count:

Sunny, Rainy, Cloudy

But what about:

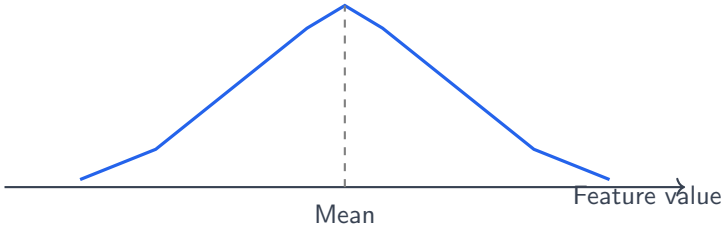
$$\text{Age} = 47.3$$

How many training examples have exactly this value?

The Solution: Learn the Shape of the Data

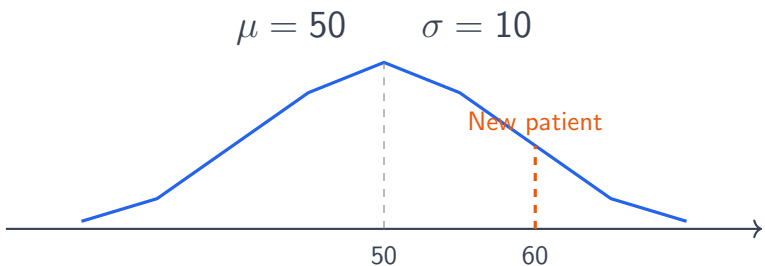
Instead of counting exact values...

Learn how values are distributed.



Example: Is Age = 60 Typical?

For patients with Disease = Yes:



Think

Is age 60 highly unusual for this class?

Gaussian Naïve Bayes

The Gaussian distribution converts a continuous value into a likelihood.

$$P(x_k = v_k \mid C_i) = \frac{1}{\sqrt{2\pi}\sigma_{C_i}} \exp\left(-\frac{(x - \mu_{C_i})^2}{2\sigma_{C_i}^2}\right)$$

Mean
μ

Spread
σ

Categorical vs Continuous Features

Categorical	Continuous
Values such as: Sunny, Rainy, Yes, No **Count occurrences**	Values such as: Age, Temperature, Income **Estimate a distribution**
Relative frequency	Gaussian likelihood

Same goal: calculate

$$P(\text{Feature value} \mid \text{Class})$$

Naïve Bayes: Why Use It? Why Be Careful?

Why it is useful

- Simple to train
- Fast prediction
- Works with small datasets
- Handles multiple features
- Effective for many classification tasks

The big limitation

Features are assumed to be conditionally independent.

But in reality:
Age $\leftrightarrow$ **Blood Pressure**
Fever $\leftrightarrow$ **Cough**

The surprising part

Even when the independence assumption is not perfectly true, Naïve Bayes can still perform well.

Live Experiment: Gaussian Naïve Bayes

```
from sklearn.naive_bayes import GaussianNB

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print(model.score(X_test, y_test))
```

Ask the class first

What information does the model learn during training?

Answer

Class priors + feature distributions for each class

Why Does Such a Naïve Model Work?

The assumption is simple.

Sometimes wrong.

But the model is:

Fast **Simple** **Effective**

Key Lesson

A model does not need to model everything perfectly to be useful.